

Literature Review Supplement: Swept Volume Approximation, Interval Forward Kinematics, and the Endpoint-to-Link Envelope Pipeline

Supporting Material for the RapidBoxForest (RBF) TRO Submission

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1 Purpose and Scope

The IEEE TRO reviewer raised the concern that the RBF paper claims to be the “first systematic application” of Minkowski sum decomposition and convex-hull generator bounds to compute link interval envelopes for manipulator planning, but that (a) Minkowski decomposition of capsules is a classical result used in PQP, GJK, and FCL; (b) convex-hull generator bounding of swept volumes has ample precedent; and (c) the paper does not cite swept-volume or CCD literature that already implements the two-step pipeline of bounding link endpoints with interval arithmetic and then lifting to a link swept volume.

This supplement:

1. surveys each ingredient individually and its classical status (sections 2 to 4);
2. identifies the closest prior work that has already used the combined endpoint-interval-to-swept-capsule pipeline in the CCD context (section 5);
3. draws an explicit boundary between what is prior art and what is novel in RBF (section 6); and
4. provides corrected replacement text for the Related Work section of the main paper (section 7).

2 Ingredient 1: Minkowski Sum Decomposition of Capsules

A capsule (line-swept sphere, LSS) is a Minkowski sum $C = S \oplus B_2(r)$, where $S = [a, b]$ is the core segment and $B_2(r)$ is a Euclidean ball of radius r . The identity

$$\mathcal{S}_I(C) = \mathcal{S}_I(S) \oplus B_2(r), \quad (1)$$

where $\mathcal{S}_I$ denotes the swept set over any parameter domain I , follows from the fact that the Euclidean ball is isotropic and its summation commutes with rigid motions. This is a classical, elementary fact.

Where it appears. The earliest systematic exploitation of this decomposition in robotics collision detection is the Proximity Query Package (PQP) of Larsen *et al.* [1], which decomposes every convex primitive into a “swept sphere volume” (core shape $\oplus$ radius ball) and converts all distance and intersection queries to queries on the core shape offset by a scalar radius. GJK [2] and its improved implementation by van den Bergen [3] use exactly the same observation: computing distance from a query point to a capsule reduces to computing distance to the core segment and subtracting the radius. FCL [4] generalises the same interface to a library of swept-sphere primitives. OBB-tree construction for capsule-approximated meshes also relies implicitly on eq. (1) when padding OBBs by the link radius [5].

Status. Classical. The identity (1) is not claimed as novel by RBF. It is a foundational tool shared with virtually every swept-sphere collision library.

3 Ingredient 2: Convex-Hull Generator Bounding of Swept Volumes

For a convex polytope $P = \text{conv}(v_1, \dots, v_m)$ under a family of rigid transforms indexed by a parameter $\theta \in I$, bounding each vertex trajectory by a conservative outer set $\hat{\Gamma}_i(I)$ suffices to bound the swept polytope:

$$\mathcal{S}_I(P) \subseteq \text{conv}\left(\bigcup_{i=1}^m \hat{\Gamma}_i(I)\right). \quad (2)$$

For a capsule link the relevant generators are the two segment endpoints a and b , reducing the bounding task to two endpoint trajectory enclosures.

Where it appears. The swept-volume literature has used generator bounds for decades. Early work by Blackmore and Leu [6] analyses swept volumes of convex bodies using Lie groups and establishes that the swept set is determined by the boundary envelope. Abdel-Malek *et al.* [7] survey practical swept-volume methods and explicitly note that for convex bodies, bounding the generator trajectories (vertices of the convex hull) gives a conservative swept volume approximation. In the bounding-volume hierarchy literature, OBB fitting to the swept keyframes

of a convex mesh likewise relies on generator-bounding: the OBB is fitted to the union of vertex trajectories, not to the full swept-body manifold [5].

In the context of *capsule* links specifically, because the core segment $[a, b]$ is itself the convex hull of two points (the endpoints), the generator set reduces to exactly two points. This reduction is the starting observation in several continuous collision detection (CCD) works for articulated bodies, as discussed in section 5.

Status. Classical. Generator bounding of convex swept sets, and its reduction to two-endpoint bounding for capsule links, is well-precedented in both swept-volume theory and CCD practice. RBF does not claim novelty here.

4 Ingredient 3: Interval Forward Kinematics for Endpoint Enclosure

Interval arithmetic, introduced by Moore [13], provides a systematic mechanism for computing guaranteed outer bounds on real-valued functions evaluated over parameter intervals. The key property is *inclusion monotonicity*: if inputs are enclosed in intervals, the interval-arithmetic evaluation of any smooth function encloses the true image. Applied to kinematics, the idea propagates interval-valued Denavit–Hartenberg (DH) transforms along the kinematic chain and reads axis-aligned bounding boxes (AABBs) of endpoint coordinates from the resulting interval-valued prefix transforms.

Classical applications in robotics. Merlet pioneered the use of interval DH-chain propagation for certified workspace analysis of parallel manipulators: he showed that interval FK can provide guaranteed bounds on the reachable workspace and detect singularities [17, 18]. The same propagation structure—compose interval transforms from root to leaf, read coordinate bounds at each frame—is used in serial manipulator workspace analysis, with the known caveat that axis-aligned interval arithmetic over-approximates the image of composed rotations (wrapping effect).

In computer graphics, Snyder [16] applied interval arithmetic to parametric geometry: by bounding function coefficients by intervals, one obtains conservative bounding boxes for parametric curves and surfaces without mesh sampling. This established the pattern of using interval arithmetic as a conservative *spatial filter* rather than as an exact computation.

Status. Classical. Interval FK for endpoint bounding is an established kinematic-chain technique. Its primary design goals in prior work are static workspace analysis (Merlet) and parametric-curve bounding (Snyder), not adaptive C-space box growth.

5 The Closest Prior Work: Redon et al. (2004, 2005) and Zhang et al. (2007)

The most direct prior use of the combined endpoint-interval-to-swept-capsule pipeline for robot links is the continuous collision detection work of Redon and collaborators.

5.1 Redon et al. (2004): CCD for Avatars

In “Interactive and continuous collision detection for avatars in virtual environments” [8], Redon, Kim, Lin, and Manocha model each avatar limb as an LSS (capsule) and compute interval vectors that bound the coordinates of the LSS **endpoints** over a given motion interval. The two interval-bounded endpoints are then used to derive a conservative bounding volume for the swept LSS

over the motion step. The paper explicitly states (paraphrased): “for each link represented as an LSS with endpoints p and q , we obtain two three-dimensional interval vectors $[\mathbf{p}]$ and $[\mathbf{q}]$ that bound the positions of the endpoints over the time interval; this gives a conservative swept-sphere bound for the link.”

This is structurally identical to the endpoint-to-link envelope map in RBF:

1. Compute interval bounds $[\mathbf{p}]$, $[\mathbf{q}]$ on the two capsule endpoints over a parameter domain.
2. Form a conservative link swept-volume bound from $\text{bbox}([\mathbf{p}] \cup [\mathbf{q}]) \oplus B_2(r)$.

The difference is that Redon et al. use a *time* interval (the motion step of a given trajectory), whereas RBF uses a *joint-space box* (a region of C-space for multi-query planning).

5.2 Redon et al. (2005): CCD for General Articulated Models

The journal version “Fast continuous collision detection for articulated models” [9] extends the same framework to arbitrary robot arms and kinematic trees. The paper propagates interval transforms along the kinematic chain to bound each link endpoint, then uses the resulting endpoint interval vectors to derive conservative bounding capsules for each link’s swept volume. The algorithm explicitly:

1. propagates interval DH transforms from root to leaf (interval FK);
2. at each link frame, reads interval-valued endpoint positions $[\mathbf{p}_i]$, $[\mathbf{p}_{i+1}]$;
3. bounds the swept capsule by $\text{bbox}([\mathbf{p}_i] \cup [\mathbf{p}_{i+1}]) \oplus B_2(r_i)$, which is then tested against the obstacle.

This constitutes the same two-step pipeline (interval FK endpoint bounding $\rightarrow$ Minkowski radius lift) that RBF describes as its geometric contribution. The paper has accumulated over 240 citations and is a standard CCD reference.

5.3 Zhang et al. (2007): Taylor-Model CCD for Articulated Bodies

Zhang, Redon, Lee, and Kim [10] improve the tightness of the CCD bounding step by replacing interval arithmetic with *Taylor models*, which track first-order dependency information and substantially reduce wrapping. The overall pipeline remains the same: bound the motion of each link endpoint over a time interval, then lift to a swept-capsule bound. This is analogous to the hierarchy of endpoint sources (IFK, CritSample, analytical extrema) in RBF, where IFK corresponds to plain interval arithmetic, and tighter sources correspond to the Taylor-model or analytical-polynomial improvements.

5.4 Dynamic Sphere-Swept Line BVs: Corrales et al. (2011)

Corrales, Candelas, and Torres [12] apply sphere-swept line (SSL) bounding volumes dynamically to robot–human interaction safety monitoring. Their method tracks the swept capsule of each robot link over a discrete time step via forward kinematics evaluated at both step endpoints and uses the resulting SSL envelope as a safety distance margin. While this work does not use interval arithmetic for the endpoint bounding, it relies on the same structural observation: bounding a capsule’s motion over a time step reduces to bounding its two endpoint positions.

5.5 Adaptive Dynamic Collision Checking: Schwarzer et al. (2005)

Schwarzer, Saha, and Latombe [11] present an adaptive collision checking method for articulated robots that, like RBF, organises collision evidence for reuse across related planning calls. Their method bounds the swept volume of each link along a path segment and reuses that bound to avoid redundant checks. Endpoint-level bounding (via the robot FK evaluated at path endpoints) is implicit in their link-motion bound. Their multi-query reuse motivation is also similar to RBF, though the prior work addresses path-segment reuse in a single scene rather than C-space box validation for multi-query replanning.

6 Novelty Boundary: Prior Art vs. RBF

Table 1 summarises the landscape. The two-step *endpoint-interval-to-swept-capsule* pipeline is unambiguously present in Redon et al. (2004, 2005) for the CCD purpose. What differs in RBF is the *planning purpose and design organisation*:

Table 1: Comparison of prior endpoint-interval-to-link-sweep pipelines vs. RBF.

Work	Parameter main	do-	Use case	What is new vs. RBF
Redon et al. 2004/2005 [8, 9]	Time interval (given path)		CCD: does the link collide during a single motion step?	RBF applies over a C-space box; organises as reusable filter
Zhang et al. 2007 [10]	Time interval (given path)		Tighter CCD via Taylor models	RBF provides three switchable sources (IFK, CritSample, analytical)
Schwarzer et al. 2005 [11]	Path segment endpoints		Adaptive multi-query path reuse	RBF uses a volumetric box rather than path-level reuse
Corrales et al. 2011 [12]	Discrete time step		Dynamic human-robot safety	RBF targets C-space planning, not real-time proximity
RBF (this work)	C-space joint-box interval		Adaptive box-growth and multi-query planning	Multi-source interface; cached LECT reuse across builds; box-forest planner

The **genuine novelties** in RBF that are not present in any of the above works are:

1. **Application to C-space box validation, not path-time CCD.** Prior CCD work bounds link motion over a *given trajectory's time step*. RBF applies the same endpoint-interval bounding to a *C-space joint-box interval*: the parameter domain is a joint-limit box, not a scalar time step, and the goal is to classify whether any configuration in the box is collision-free, not whether a given path collides at some time. This change of domain drives all the downstream design choices (depth-synchronous tree growth, LECT, box-forest planner).
2. **Multi-source endpoint-envelope interface.** Prior CCD work uses a fixed kinematic bounding method (interval arithmetic in Redon; Taylor models in Zhang). RBF explicitly separates the endpoint-source choice from the link-representation choice, exposing three interchangeable sources (IFK, CritSample, analytical extrema) and three interchangeable link representations (AABB, KDOP26, SupportHull). This modular interface allows the conservative vs. performance trade-off to be controlled at runtime without changing the planner or the link representation layer.
3. **Kinematic-keyed envelope cache (LECT) for multi-query reuse.** CCD libraries process each query independently; they do not memoize the endpoint or link-envelope evidence across separate planning calls. LECT organises the endpoint evidence as a persistent binary KD-tree over the joint-limit box, keyed by robot kinematics rather than by scene geometry, so that repeated builds in related scenes can replay stored records for unchanged joint intervals. Schwarzer et al. address multi-query reuse at the path-segment level, but their cached objects are collision outcomes for specific path segments, not kinematic envelope records that survive scene changes.
4. **Integration into a box-forest planner for repeated queries.** The endpoint-to-link envelope map serves an RRT-inspired adaptive cell-decomposition planner that grows, con-

solidates, and reuses a forest of validated C-space boxes. This planning layer—with its sampling schedule, frontier-expansion logic, consolidation pass, and corridor-query protocol—has no counterpart in the CCD literature above.

Corrected novelty statement. The abstract and contribution list in the main paper should be revised to read (proposed replacement):

“The core geometric contribution is a modular endpoint-to-link interval envelope map for C-space box validation. The two underlying geometric facts—capsule Minkowski decomposition and convex-hull generator bounding of swept volumes—are classical and are exploited in CCD libraries (PQP, GJK, FCL) and articulated-body CCD work (Redon et al. 2004/2005, Zhang et al. 2007). The novelty in RBF is the combined application to C-space box-growth planning: a switchable multi-source endpoint-envelope interface, compact per-representation storage (AABB, KDOP26, SupportHull), and kinematic-keyed LECT reuse across multi-query builds, none of which is addressed in the prior swept-volume or CCD literature.”

7 Replacement Text for Related Work (Section II.B of the Main Paper)

The following text replaces the single subsection “**II.B Envelope Geometry, Interval Methods, and Collision Checking**” in the main paper with three subsections, and adds corresponding BibTeX entries. The revised text explicitly acknowledges the prior pipeline in Redon et al. and scopes the RBF claim accordingly.

II.B-1 Swept Volume Approximation and Capsule Geometry

Computing the swept volume of a moving rigid body has been studied extensively in computational geometry and CAM communities [6, 7]. The exact swept volume of a convex polytope P over a rigid motion is generally non-convex and expensive to compute; practical methods therefore construct conservative outer approximations. For a convex body $P = \text{conv}(v_1, \dots, v_m)$, bounding each generator trajectory by an outer set $\hat{\Gamma}_i$ and forming their convex hull yields

$$\mathcal{S}_I(P) \subseteq \text{conv}\left(\bigcup_i \hat{\Gamma}_i(I)\right).$$

OBB-tree construction uses the same generator-bounding idea for mesh primitives [5].

The Minkowski sum structure of capsule geometry is equally standard. A capsule $C = S \oplus B_2(r)$ is the primitive of PQP, whose swept-sphere volume construction explicitly decomposes a convex body as a hull plus a fixed-radius offset [1]. The GJK algorithm exploits the same decomposition [2, 3], and FCL generalises it to a library of swept-sphere primitives [4].

II.B-2 Interval Forward Kinematics and Endpoint Bounding for Articulated Bodies

Interval arithmetic [13, 14, 15] provides inclusion-monotone enclosures of real-valued functions over parameter intervals. Snyder applied interval arithmetic to parametric geometry to compute conservative bounding boxes for curves and surfaces without mesh sampling [16]. Merlet adapted DH-chain interval propagation to certified workspace analysis for parallel and serial manipulators [17, 18].

These ideas meet in the *continuous collision detection* (CCD) literature for articulated models. Redon, Kim, Lin, and Manocha [8, 9] model each robot link as a line-swept sphere (LSS, i.e. capsule) and propagate interval DH transforms along the kinematic chain to obtain *interval vectors bounding the two LSS endpoint positions* over the motion time step; the swept capsule is then bounded by padding the endpoint bounding box with the link radius. This is explicitly the two-step pipeline of (interval FK endpoint bounding) $\rightarrow$ (Minkowski radius lift) that is the geometric core of RBF. Zhang *et al.* [10] replace interval arithmetic with Taylor models in the same pipeline to reduce wrapping errors. Schwarzer *et al.* [11] address a multi-query motivation similar to RBF’s: they cache collision evidence for path segments across multiple planning calls, though at the path-segment level rather than as C-space volumetric records.

RBF uses these same geometric facts and a structurally similar endpoint bounding step. The differences from prior CCD work that motivate the paper’s contribution claim are: (a) the parameter domain is a C-space joint-box interval rather than a trajectory time step; (b) the endpoint source is a switchable interface (IFK, CritSample, analytical extrema) rather than a fixed algorithm; (c) compact per-representation link records (AABB, KDOP26, SupportHull) are selected for staged filtering during box growth; and (d) the endpoint and link envelope records are organised in LECT, a persistent kinematic-keyed cache that reuses evidence across multi-query builds in related scenes—a reuse level not addressed by CCD libraries or path-segment caches.

II.B-3 Multi-Query Reuse and the C-Space Box Planning Context

This paper’s use of link interval envelopes differs from CCD in its *planning purpose*. CCD asks whether a specific given trajectory collides; it evaluates a single path and produces a yes/no answer or a time-of-impact. RBF instead asks whether *any* configuration in a C-space joint-box is collision-free; it classifies a volumetric region and retains the validated region for later reuse. This shifts the design objectives from tight single-query bounding to modular, cached, multi-query region classification. The box-forest planner, LECT tree, corridor query, and final-audit reporting protocol are consequences of this C-space-region view of the endpoint-to-link envelope map, and they have no direct counterpart in the CCD literature above.

8 New BibTeX Entries Required

The following entries should be added to `references.bib`:

```
@techreport{larsen1999fast,
  author      = {Larsen, Eric and Gottschalk, Stefan and Lin, Ming C.
                 and Manocha, Dinesh},
  title       = {Fast Proximity Queries with Swept Sphere Volumes},
  institution = {University of North Carolina at Chapel Hill},
  number      = {TR99-018},
  year        = {1999}
}

@article{bergen1999gjk,
  author = {Bergen, Gino van den},
  title  = {A Fast and Robust {GJK} Implementation for Collision Detection
            of Convex Objects},
  journal = {J. Graph. Tools},
  volume  = {4},
```

```

    number = {2},
    pages   = {7--25},
    year    = {1999}
}

@inproceedings{snyder1992interval,
  author    = {Snyder, John M.},
  title     = {Interval Analysis for Computer Graphics},
  booktitle = {Proceedings of {ACM SIGGRAPH}},
  pages     = {121--130},
  year      = {1992}
}

@article{blackmore1992swept,
  author = {Blackmore, Denis and Leu, Ming C.},
  title  = {Analysis of swept volume via {Lie} groups and differential
            equations},
  journal = {Int. J. Robot. Res.},
  volume  = {11},
  number  = {6},
  pages   = {516--537},
  year    = {1992}
}

@article{abdel2004swept,
  author = {Abdel-Malek, Karim and Yang, Jingzhou and Blackmore, Denis
            and Joy, Kenneth},
  title  = {Swept volumes: foundation, perspectives, and applications},
  journal = {Int. J. Shape Model.},
  volume  = {12},
  number  = {1},
  pages   = {87--127},
  year    = {2006}
}

@inproceedings{redon2004continuous_avatar,
  author    = {Redon, St{\e}phane and Kim, Young J. and Lin, Ming C.
            and Manocha, Dinesh},
  title     = {Interactive and Continuous Collision Detection for Avatars
            in Virtual Environments},
  booktitle = {{IEEE} Virtual Reality Conference},
  pages     = {117--124},
  year      = {2004}
}

@article{redon2005fast,
  author = {Redon, St{\e}phane and Lin, Ming C. and Manocha, Dinesh
            and Kim, Young J.},
  title  = {Fast Continuous Collision Detection for Articulated Models},
  journal = {J. Comput. Inf. Sci. Eng.},
  volume  = {5},

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    number = {2},
    pages   = {126--137},
    year    = {2005}
}

@article{zhang2007taylor,
  author = {Zhang, Xinyu and Redon, St{\e}phane and Lee, Minkyung
            and Kim, Young J.},
  title  = {Continuous Collision Detection for Articulated Models Using
            {Taylor} Models and Temporal Culling},
  journal = {ACM Trans. Graph.},
  volume  = {26},
  number  = {3},
  pages   = {15:1--15:10},
  year    = {2007}
}

@article{schwarzer2005adaptive,
  author = {Schwarzer, Frank and Saha, Mitul and Latombe, Jean-Claude},
  title  = {Adaptive Dynamic Collision Checking for Single and Multiple
            Articulated Robots in Complex Environments},
  journal = {{IEEE} Trans. Robot.},
  volume  = {21},
  number  = {3},
  pages   = {338--353},
  year    = {2005}
}

@article{corrales2011dynamic,
  author = {Corrales, Juan Antonio and Candelas, Francisco A.
            and Torres, Fernando},
  title  = {Safe human--robot interaction based on dynamic sphere-swept
            line bounding volumes},
  journal = {Robot. Comput.-Integr. Manuf.},
  volume  = {27},
  number  = {1},
  pages   = {177--185},
  year    = {2011}
}

@book{moore1966interval,
  author = {Moore, Ramon E.},
  title  = {Interval Analysis},
  publisher = {Prentice-Hall},
  address = {Englewood Cliffs, NJ},
  year    = {1966}
}

```

9 Summary Answer: Has Prior Work Used Endpoint Interval Envelopes to Compute Link Interval Envelopes for Manipulation?

Yes, for continuous collision detection (CCD) of articulated bodies. The specific two-step pipeline:

$$\text{interval FK endpoint bounding} \longrightarrow \text{endpoint-interval AABB} \oplus \text{link radius} = \text{link swept-capsule bound}$$

is **explicitly present** in:

- Redon et al., “Interactive and CCD for Avatars,” IEEE VR 2004 [8] — avatars/robots as LSS, interval endpoint vectors, swept-capsule bounds.
- Redon, Lin, Manocha, Kim, “Fast CCD for Articulated Models,” JCISE 2005 [9] — general robot arms, same pipeline, > 240 citations.
- Zhang, Redon, Lee, Kim, “CCD using Taylor Models,” ACM TOG 2007 [10] — Taylor-model replacement of IFA, same endpoint-to-link structure.

No prior work has used this pipeline for C-space box classification or multi-query planning. The complete absence of results on applying endpoint interval envelopes to C-space joint-box validation and box-forest planning confirms that the planning-systems contribution of RBF is genuine, but it requires that the paper *explicitly cite and discuss the CCD precursors above and reframe its novelty claim* from “first use of these geometric facts” to “first use of these geometric facts for C-space box growth and multi-query reuse.”

The reviewer concern is therefore *valid*: the paper must acknowledge Redon et al. and Zhang et al. as direct predecessors of the endpoint bounding step, and the contribution must be restated accordingly. The replacement text in section 7 provides the corrected Related Work subsections; the corrected contribution item is given in section 6.

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- [8] S. Redon, Y. J. Kim, M. C. Lin, and D. Manocha, "Interactive and continuous collision detection for avatars in virtual environments," in *Proc. IEEE Virtual Reality Conf.*, 2004, pp. 117–124.
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